

What's Cooking? Olfactory Sensing Using Off-the-Shelf Components

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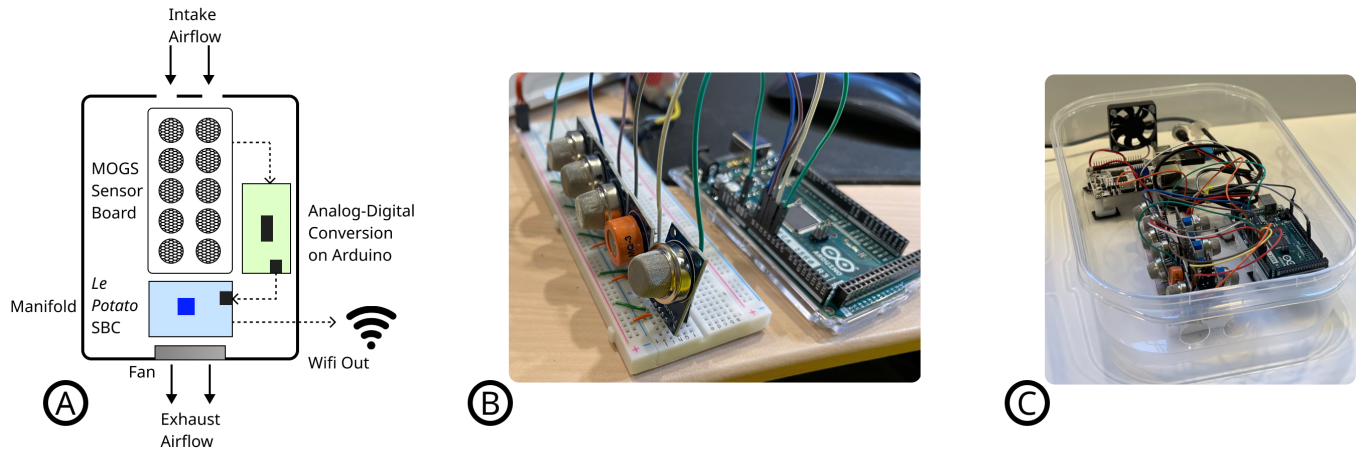


Figure 1: (a) Hardware schematic showing layout of components and airflow. (b) Benchtop prototype using Arduino. (c) Assembled prototype including fan and single-board computer integrated in manifold (with lid removed).

ABSTRACT

We present *Project Sniff*, a hardware and software exploration into olfactory sensing with an application in digital communication and social presence. Our initial results indicate that a simple hardware design using off-the-shelf sensors and the application of supervised learning to the sensor data allows us to detect several common household scents reliably. As part of this exploration, we developed a scent-sensing IoT prototype and placed it in the kitchen area to sense “what’s cooking?”, and share the olfactory information via a Slack bot. We conclude by outlining our plans for future steps and potential applications of this research.

KEYWORDS

olfactory sensing, inference, immersive communication, AR, VR

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1 INTRODUCTION

Traditional messaging has focused on text, audio, image or video as the medium of choice for communication. Transmitting of olfactory sensations has been less widely adopted. However, with the rise of immersive communication [12], the olfactory channel could potentially take on a more important role. Olfactory sensations have the potential to shape the way we perceive certain scenes or situation in a meaningful and fundamental way [1, 5] – creating multi-sensorial experiences.

Previous work has explored olfactory output devices [6, 7] for interactive applications. Using olfactory output devices, work on olfactory messaging has shown that olfactory messages reduce disruptiveness to the user [3] and can increase the user’s awareness of message urgency [9]. However, in order to more directly convey information about the sender’s location or information specific to the contents of the message, we believe, in particular for immersive messaging (e.g., as a side-channel for AR/VR messages), that it would be highly useful to have a device that is able to detect scents in the environment of the user, such that they can be re-transmitted in messages.

In this poster, we present *Project Sniff*, which explores the usage of olfactory information for digital communication and social presence. Our olfactory input prototype is able to detect multiple scents commonly found in a household setting. We describe our

Table 1: The types and response properties of the sensors used in our prototype.

Sensor Type	Primary Compounds with Sensor Response
MQ-2	methane, butane, LPG, smoke
MQ-3	alcohol, ethanol, smoke
MQ-4	methane gas / CNG
MQ-5	natural gas, LPG
MQ-6	LPG, iso-butane, propane
MQ-9	carbon monoxide, combustible gasses
MQ-135	benzene, alcohol, smoke
MQ-136	hydrogen sulfide gas
MQ-137	ammonia
MQ-138	benzene, toluene, alcohol, acetone, propane, formaldehyde gas, hydrogen

prototype’s hardware and software design, outline our deployment and potential applications, followed by future work.

2 SCENT SENSING HARDWARE

The human olfactory system uses about 300 olfactory receptors to sample scents inhaled through the nose. Individual human olfactory receptors are sensitive to multiple chemical compounds present in the inhaled scent. These receptors’ responses are transmitted to the brain, which combines and processes the “signals” of the olfactory receptors to determine the scent [11].

Our hardware design mimics biological olfactory sensing by using a sampling manifold, a fan to draw in air from the outside and an array of off-the shelf gas sensors, each with different response curves to a number of compounds present in the air. Figure 1 (a) shows a schematic of our prototype’s design. We chose to use “MQ Series” metal oxide semiconductor (MOS) sensors, as they are well-characterized [8], cheap, widely available and, in contrast to quartz crystal microbalance (QCM) sensors are less sensitive to interference from environmental conditions such as airflow effects and electromagnetic effects from other QCM sensors operating in the vicinity [2]. An advantage of using QCM sensors would be the ability to detect more complex molecules than possible for MOS sensors, such as specific proteins. Table 1 shows the sensors we used in our prototype and the specific compounds they are able to respond to.

We use the analog–digital converter of an *Arduino Mega 2560*¹ to convert the outputs from the sensor boards. This information is then transmitted via a serial connection to a *Libre Computer Le Potato*² single board computer (SBC) for further processing.

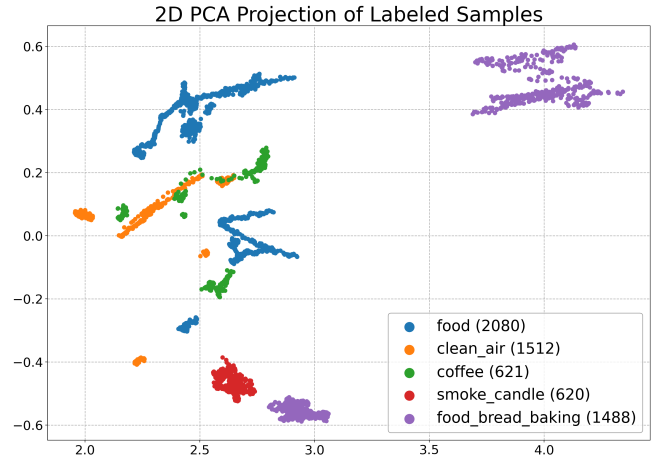
3 SCENT CLASSIFICATION MODEL

We treat scent recognition as a supervised learning problem. As a proof of concept we selected the following scents commonly found in a household or a kitchen area as classes for classification: (1) *clean air* (no scent), (2) *food* (food and cooking smells), (3) *coffee* (smell emitted while brewing coffee) and (4) *bread baking*. To obtain training examples for supervised learning with our prototype, we

Table 2: Scent Classification Results.

Metric	SVM (test)	SVM (valid)	DT (test)	DT (valid)
Precision	0.979	0.976	0.985	0.981
Recall	0.979	0.978	0.986	0.981
F1	0.979	0.975	0.985	0.981
Accuracy	0.978	0.974	0.985	0.981

wrote a logging application running on the SBC that samples the sensor voltages in one second intervals, and writes them to a log file together with a label. We captured over 600s of scent samples for each class.


Figure 2: 2D PCA projection of labeled samples. Non-significant overlap of samples indicates good separability of the scent classes.

To verify if the various scents are distinguishable, we projected the labeled sample points into two dimensions via Principal Component Analysis (PCA). As can be seen in Figure 2, the scent classes have relatively little overlap. This indicates that scent classes are (at least) visually distinguishable, and that supervised learning techniques are applicable to the data set.

To evaluate classification accuracy, we trained a linear support vector machine (SVM) classifier and a decision tree (DT) classifier with 620 samples per scent class. For the DT classifier we used the Gini criterion and set a minimum of 5 samples per leaf and minimal cost-complexity pruning [4] coefficient of 0.1 to mitigate over fitting. We used the classifier implementations from the Python SKLearn package [10]. We used a 75 % : 15 % : 10 % train–test–validation split. Table 2 shows our classification results. Overall the results seem to be consistently good in the high 90 % range, with a very slight improvement of DT over SVM.

4 PROTOTYPE DEPLOYMENT AND FUTURE WORK

Due to our current prototype’s size (36 × 24 × 9.5 cm), we designed our demo application around the idea of using the prototype as

¹<https://docs.arduino.cc/hardware/mega-2560>

²<https://libre.computer/products/s905x/>

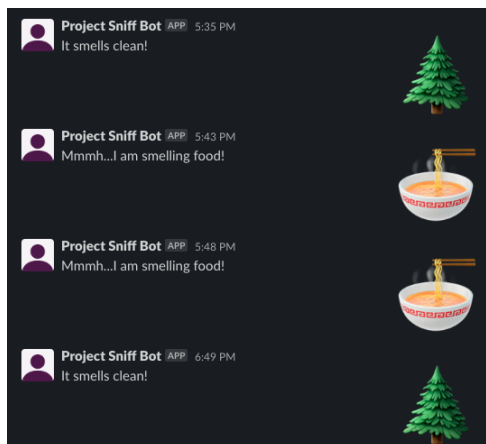


Figure 3: Sample output of *Project Sniff Bot* classifications over time.

a scent-sensing IOT appliance, that is always on and continuously classifies the current scent in a space. Therefore, we implemented a server that is started automatically with the SBC startup which connects to and reads from the sensor array running on the Arduino. Using the DT scent classification model discussed in the previous section, this server provides a web endpoint that displays the currently detected and history of recently detected scents. Furthermore, we implemented a simple Slack Bot (*Project Sniff Bot*), that communicates the currently detected scent to a Slack channel every hour and on every change of detected scent. Such updates can augment the awareness of what is happening in the environment of a remote co-worker without being as intrusive as, e.g., a camera feed — enabling social presence. So far, this prototype has generated positive feedback within our team. Figure 3 shows a screenshot of some of the Sniff Bot’s updates.

Our prototype demonstrated that it is possible to enable olfactory sensing using off-the-shelf components. By capturing an additional sensory information — smell — in addition to visuals and audio, novel and richer forms of communication systems can emerge. Besides our prototype with Slack bot, we imagine potential applications to use olfactory transducers on the receiver’s side to allow transmission of olfactory sensations from the sender’s environment to the receiver. For instance, by augmenting an existing AR messenger application with *Olfactory Stickers*, a user can send olfactory information with an AR object to their friends. Figure 4 shows an example of the “olfactory sticker” concept. The olfactory sticker could encode the scent of the object being send (e.g., a pineapple) to the user. We use the sticker metaphor to enable the viewer to preview the scent, but only release it when they are ready, e.g., by rubbing the sticker on their device’s touch screen or head-worn AR/VR device.

In the future, we aim to increase the number of detectable scents and, consequently, to better understand the limitations of MOS sensor arrays for scent detection. Having obtained a better characterization of these sensors, we would furthermore be interested in developing custom hardware to contain a miniaturized MOS sensor array, geared specifically towards scent detection on smart phones

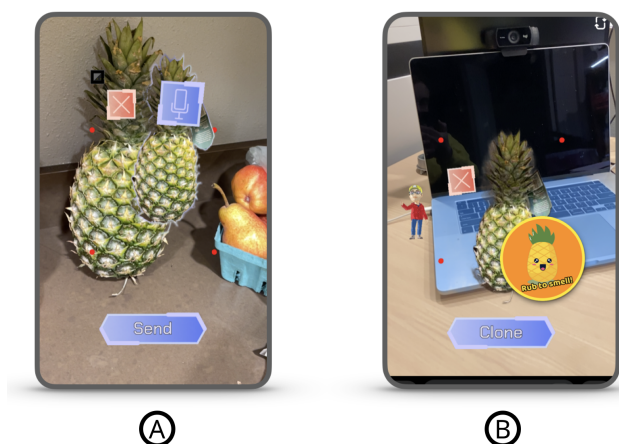


Figure 4: Mock-up of an app that allows the transmission of olfactory information via an *olfactory sticker*. (a) Pineapple is “cloned” into AR, and its scent is detected. (b) Receiver receives pineapple clone, and an olfactory sticker representing the scent. By rubbing on it, the receiver can activate the olfactory sticker’s scent.

or wearable AR/VR devices for multi-sensorial communication experiences.

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